

Logs from May 19 to the 25th

1 Meeting May 19

1.1 Summary of last week's work:

1. The repository is now up and running, with meeting logs and daily progress logs.
2. Last week, I focused mainly on reviewing functional analysis. This included:
 - reviewing linear, bounded, closed, compact, and inverse operators;
 - reviewing Riesz's lemma, the Open mapping theorem, the Closed graph theorem, the Bounded Inverse theorem, Neumann series, Riesz's representation theorem, and the Riesz map.
 - studying the spectrum of a linear operator, including the distinction between point, continuous, and residual spectra;
 - learning about Banach and Hilbert adjoints from Umberto's first lecture;
 - comparing self-adjoint, unitary, and normal operators;
 - learning about the Fredholm alternative for compact linear operators from Umberto's first lecture.
3. I also went through the first chapter of Nick's text. This included:
 - the general motivation for computing eigenvalues;
 - the motivation for computing pseudospectra of nonnormal operators;
 - eigenvalues of matrices and linear operators;
 - pseudospectra of matrices;
 - pseudospectra of linear operators.

4. To further consolidate this material, I went through Umberto's first two lectures. Here, I learned about:
 - Hilbert, Banach, and formal adjoints;
 - spectra and pseudospectra of linear operators;
 - spectral radii of bounded operators;
 - properties of spectra for different classes of operators. For example, the spectrum of a compact operator consists of isolated points.
 - separable Hilbert spaces and Hilbert bases;
 - the spectral theorem for operators that are compact and self-adjoint or compact and normal.
5. I went through the preliminary section of Matt's text to fill in any remaining gaps. Here, I learned more about:
 - separable Hilbert spaces and orthogonal projections;
 - the Uniform Boundedness Theorem;
 - Matt's definition of the adjoint;
 - boundedly invertible operators;
 - bounded invertibility of closed and bounded operators;
 - Matt's definition of the spectrum, and how it compares with the classical functional analysis definition in Kreyszig's text;
 - spectra of infinite direct sums of operators;
 - spectra of closed and densely defined operators;
 - symmetric, self-adjoint, and essentially self-adjoint operators.
6. Finally, I started to type up some of my handwritten notes on what I had learned. I think this is helping me connect and consolidate the material. I want to continue this habit!

1.2 Meeting Notes

- In the repository, I should upload the `.tex` files instead of the compiled PDFs.
- In my background notes, I should cite specific theorems using a format such as `\cite[Theorem 1]{person}`.

- After finishing my written notes, I should move on to the finite element discretization of eigenproblems. The main source for this will be Boffi's text.
- I should also start working through examples of eigensolves in Firedrake, possibly using tutorials. Eventually, I should work through the curl-curl example to see the potential issues that arise when computing spectra.
- After learning about these discretizations, it may be a good time to go through the literature on multilevel solvers for eigenvalue problems. It would be useful to understand the current state of the art.
- Firedrake assembles PETSc vectors and matrices, and uses PETSc's KSP objects for linear solves and SNES objects for nonlinear solves. On top of PETSc is SLEPc, the "Scalable Library for Eigenvalue Problem Computations." SLEPc adds an EPS object for eigenproblems. Thus, Firedrake's linear eigensolve creates an EPS object. The SLEPc manual and technical notes are useful resources for learning about eigensolves and their implementation.
- The next meeting is Monday, May 25, at 3 p.m. UK time.
- I have a meeting with Matt Colbrook sometime on June 11.
- The presentation is in the first week of June. I will start thinking about what this should look like soon.

1.3 This Week's Plan

1. Today (May 19) and tomorrow, I will work on finishing up my background notes. I hope to get this done!
2. I will then move on to learning about finite element discretizations of eigenproblems. This will involve some general googling, Boffi's text, and maybe the SLEPc manual.
3. Work through some examples in Firedrake. First, I will work through some tutorials. I hope to work up to the curl-curl example.
4. Transition to a literature review for Multilevel eigensolves.

2 Daily Logs

Tuesday

- Spent the morning prepping for the meeting with Patrick, having the meeting with Patrick, and creating a new meeting log.
- I then returned to my typed-up background notes. I made the citation changes suggested by Patrick. I then proceeded to start adding in the proofs I had left out. I will continue with this tomorrow and the next day.

Wednesday

- Today was a travel day. Unfortunately, between commuting, flights, etc, not much got done. For a couple of hours on the plane I worked on typing up some proofs.

Thursday

- Today was my family's event. This meant that I only got about five-ish hours of work done. I proceeded to finish up some typed up proofs.
- I will move on to the finite element background tomorrow!

Friday

- I spent the first two hours cleaning up some final proofs in my written notes.
- Then I moved on to Boffi's text. I did a little FEM and Firedrake review to set the scene. Then I proceeded to write notes on the the simplest example (1D Laplace) and coded it up in Firedrake.

Sunday

- Working on 2D Laplace examples today.
- Took a while to figure out how to create unstructured meshes. used Chat to help. Not sure if I should use netgen or gmsh for the unstructured square meshes. Not sure how to pick a desired number of DOF in netgen. How did Boffi construct their meshes?

- Worked through some of the Laplace eigenvalue examples in 2D. In particular, I worked on unstructured meshes, structured meshes, meshes with reverse diagonals, meshes with criss-crosses, and unstructured L-shaped meshes.
- I then did a quick review on conforming vs. nonconforming elements and recreated the Boffi example with Crouzeix-Raviart elements. As expected, the eigenvalues are now approximated from below.
- Tomorrow (and the next day), I plan to work through the examples for the mixed formulation of the Laplace eigenvalue problem. Then, finally, I will move on to the Maxwell problem.
- What sections of Boffi should I work through and study? All of them?
- At some point, I want to spend some time looking through SLEPc to try and understand the solvers some more.